

Multiplicity Control vs Replication:
Making an Obvious Choice Even More Obvious

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Abstract

This paper contrasts and compares multiplicity control procedures and replication in the control of familywise error and the maximization of statistical power. Results of a simulation study are used to show that replication provides similar familywise error control and power, and is logically superior to, multiplicity control procedures.

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It is easier than ever to collect and analyze vast amounts of data. Plentiful research participants, accessible statistical software, and the popularity of the social sciences have led to a golden age of quantitative research. As the number of statistical tests being conducted has risen, social science stakeholders have become increasingly concerned with Type I errors (false positives), that is, finding a “significant” effect simply as a result of sampling error. This is because as more and more tests are conducted the probability of a Type I error occurring increases. Understandably, there have been repeated calls for the adoption of multiplicity control to reduce the number of false positive results in research (Alibrandi, 2017). At the same time, the value of replication is being touted across many disciplines as a way of ensuring that the results of scientific studies are legitimate (Cumming, 2014; Shrout & Rodgers, 2018).

To date, multiplicity control and replication have rarely been discussed within the same context. This is surprising since they both purport to reduce the likelihood of Type I errors in the results of research studies. In this paper, we discuss the tenets and principles of multiplicity control and replication, and then we move into a comparison of the methods both theoretically and methodologically.

The Multiplicity Problem

Multiplicity refers to testing multiple hypotheses with the goal of isolating those that are statistically significant. The problem is that as the number of tests conducted increases, the probability of obtaining a Type I error also increases. To illustrate this principle, let’s say we are comparing the speed at which participants walk. Participants are separated into four groups and each group is primed with a different list of words. If we hypothesize that priming will affect subsequent walking speeds, then we may wish to compare each group to every other group

individually (i.e., test all six pairwise comparisons). Though each test carries a specific probability of making a Type I error (α), the overall probability of a Type I error (α') across all six tests will be higher than α . In this way we can see how researchers are often put in the agonizing position of having interesting results that likely contain one or more false positives. Furthermore, if they cannot determine *which* results are false positives, their entire study may be received with scepticism.

Multiplicity Control Procedures

Researchers have traditionally attempted to control for the increased likelihood of a Type I error when multiple tests are conducted by using multiplicity control procedures (MCPs). There are many different MCPs, but all accomplish essentially the same goal—they make the cut-off demarcating statistically significant from statistically non-significant results more conservative as the number of statistical tests conducted increases (Olejnik, Li, Supattathum, & Huberty, 1997, Sakai, 2018). MCPs can be applied to many kinds of tests, such as pairwise mean comparisons, multiple tests of correlation or regression coefficients, multiple parameters in structural equation modeling, tests repeated over multiple outcome variables, multiple voxel-level tests in fMRI, and more.

Some of the most popular MCPs provide familywise error control (α_{FW}), which controls the probability of at least one Type I error at α across all comparisons (i.e., $\alpha' = \alpha$). The most popular approach for α_{FW} control is the Bonferroni method (Dunn, 1961), which controls for multiplicity by dividing the overall probability of a Type I error (α_{FW}) by the number of tests conducted (T). The resulting per-test alpha level is $\alpha_T = \alpha / T$. Numerous alternatives to the Bonferroni procedure for controlling α' at α have been proposed, with the Holm (1979) procedure a flexible and popular alternative. The Holm procedure makes inferences regarding

statistical significance in a stepwise manner. The term *stepwise* implies that the significance tests take place in a prespecified order and α_T can depend on the specific stage of testing.

Replication

Replication lends validity and generalizability to empirical results, and as such, has been heralded as a cornerstone of the so-called “new statistics” (Cumming, 2014). It also happens to address the multiplicity problem by leveraging the simple principle that it is highly unlikely that sampling error would yield the *same* false positive result across several studies. These are some of the reasons why direct replications are gaining traction. Indeed, many academic journals have stated that they are now open to accepting replication studies (e.g., Lucas & Donnellan, 2013; Vazire, 2016). It is our position that replication, an indispensable tool in its own right, naturally and effectively deals with the multiplicity problem.

Multiplicity Control Procedures vs. Replication

Although MCPs and replication are theoretically very different, they share a common goal in reducing the probability of Type I errors; hence we find a comparison of these strategies informative. Below we outline the reasons why we find replication to be a more logical and natural way to control for Type I errors than adopting MCPs.

To begin, there are important theoretical issues with the general practice of multiplicity control, highlighted by the fact that there is no basis for the decision to link the alpha (α) level (or maximum acceptable Type I error rate) used for a particular test to the number of *other* tests conducted within a study (Carmer & Walker, 1985; Cribbie, 2017; Rothman, 1990; Saville, 1990, 2003, 2015). Although many have highlighted that α' can increase drastically when researchers ignore the effects of multiplicity (e.g., Bland & Altma, 1995; Hancock & Klockars, 1996; Holland, Basu & Sun, 2010; Ryan, 1959; Tyler, Normand, & Horton, 2011), there is still no logical theoretical basis for linking the number of tests conducted to the per-test Type I error

rate. For example, conducting all $T = 6$ pairwise comparisons in one study is strategically no different than conducting six studies each with $T = 1$ pairwise comparison, so why should there be a penalty for conducting all the tests together? If you believe that these situations are equivalent, and that MCPs should be applied in both of these cases, why not control for all tests conducted by the researcher over their scholarly career? Or even further, all statistical tests ever conducted? The ridiculousness of this suggestion speaks to the way that the logic, or lack thereof, underlying MCPs does not scale well. In addition, linking the number of tests conducted to the per-test Type I error rate leads to strange recommendations such as limiting the number of variables studied in order to reduce the potential for Type I errors (Schochet, 2007).

Second, replication encourages researchers to repeat methods under slightly different conditions (e.g., different cities, lab settings, research assistants, samples). MCPs only address the likelihood of erroneous results within the study at hand. In contrast, replication reduces error by increasing the scope of an initial study, which directly contributes to the generalizability of the findings (Fisher, 1935; Lindsay & Ehrenberg, 1993). Repeated findings and generalizability—rather than a low chance of error in a single study—have widely been regarded as the hallmark of legitimate results (Carver, 1993; Fisher, 1935; Lykken, 1968; Nuzzo, 2014; Popper, 1934; Steiger, 1990).

Third, it has been noted that most (if not all) variables investigated in meaningful studies are related, although the magnitude of the association might not be large (Gelman, Hill, & Yajima, 2012; Rothman, 2014). Thus, procedures aimed at reducing Type I errors are unnecessary and irrelevant since true Type I errors are virtually nonexistent. To word this differently, MCPs have the single goal of reducing the likelihood of Type I errors, while replication has the broader goal of exploring the generalizability of research findings (including

the direction and magnitude of effects). If Type I errors are not an issue, then MCPs are unnecessary.

Fourth, since replication, unlike MCPs, is not a procedure founded within null hypothesis significance testing, there are many extensions that are available such as focusing on the magnitude of effect sizes and meta-analyzing the effects across the multiple replication studies.

Lastly, across popular testing situations, multiplicity control is not superior to replication in terms of reducing the likelihood of Type I errors. This novel finding is the primary focus of this paper. If replication is theoretically superior to multiplicity control while providing the statistical benefits of MCPs then there would appear to be a clear winner.

Methodology

We conducted a Monte Carlo simulation to evaluate whether replication provides a similar level of Type I error control to multiplicity control. We simulated pairwise comparisons within a one-way independent groups framework, comparing familywise error control and statistical power of the Bonferroni and Holm MCPs to that of replication.

Several factors were manipulated in this study, including sample size, number of replications, number of populations, population mean configuration, and the method of error control. Sample sizes per group were set at $n = 25$ and $n = 100$ to reflect common sample sizes in psychological research. The number of groups was set at either $J = 4$ (six pairwise comparisons) or $J = 7$ (21 pairwise comparisons).

To test both familywise error control and statistical power, we manipulated population means so that three types of configurations were adopted: all population means equal (complete null), some of the population means equal (partial null), and none of the population means equal (complete non-null). In the complete null case, we investigated familywise error rates. In the partial null case, we evaluated both familywise error rates and power. In the complete non-null

case, we investigated power. Power was recorded in terms of all-pairs power (the proportion of simulations in which all null hypotheses associated with non-null pairwise comparisons were correctly rejected) and average per-pair power (the proportion of truly significant differences that were correctly identified as statistically significant, averaged across all simulations). The population mean configurations used in the study can be found in Table 1. The within-group standard deviation was fixed at 20, and thus every one-unit increase in differences between means increased the effect size (Cohen's d) by .05. For example, the population value for Cohen's d for the $\mu_1 = 0$ and $\mu_2 = 16$ comparison is $-.8 [(0-16)/20]$. Unequal population means were equally spaced.

Familywise error control and power were evaluated under situations in which the study was not replicated, replicated once, or replicated twice. With no replication, the familywise error rate was calculated as the proportion of simulations in which at least one pairwise comparison was falsely declared statistically significant (i.e., there was at least one false positive). With one or two replications, the familywise error rate was calculated as the proportion of simulations in which at least one pairwise comparison was falsely declared statistically significant in the original study *and* in each replication (i.e., the false positive persisted across replications). With no replication, the per-pair power rate was calculated as the average proportion of simulations in which non-null pairwise comparisons were correctly declared statistically significant, whereas the all-pairs power rate was the proportion of simulations in which all non-null pairwise comparisons were correctly declared statistically significant. With replication, the per-pair power rate was calculated as the average proportion of simulations in which non-null pairwise comparisons were correctly declared statistically significant in the original study and in each replication, whereas the all-pairs power rate was the proportion of simulations in which all non-

null pairwise comparisons were correctly declared statistically significant in the original study and in each replication.

To examine familywise error rates across simulations, we adopted three methods of multiplicity control. The first method evaluated each pairwise comparison at α (i.e., no multiplicity control), the second was the Bonferroni MCP method, and the third was the stepwise Holm MCP method. In addition to computing the test statistics separately for the original study and each replication, to model the accumulation of research over time, Type I error and power rates were also investigated when the combined (meta-analytic) effect across replications was analyzed. Meta-analysis is a useful tool for combining research that examines the same effect, and here, we use it to model how replication effects may be combined.

Lastly, beyond traditional NHST-based approaches, rates were also calculated for instances in which the effect size (Cohen's d) meets the minimum meaningful value (ϵ) in both non-replicated and replicated situations. The effect size equivalent of a Type I error occurs when the observed d value mistakenly exceeds ϵ (i.e., the population value of $d < \epsilon$, but the observed value of $d > \epsilon$), whereas the equivalent of power occurs when the observed d value correctly exceeds ϵ (i.e., the population value of $d > \epsilon$, and the observed value of $d > \epsilon$). When the population value of $d > \epsilon$, we can calculate the average proportion of correct statements regarding d or the proportion of all correct statements regarding d (i.e., all pairwise d values that are greater than ϵ when population $d > \epsilon$).

For this study, the nominal Type I error rate was set at $\alpha = .05$, ϵ was set at $d = .3$, and 5000 simulations were conducted for each condition. It should be noted though that the choice of an appropriate value for α and ϵ is affected not only by general recommendations but also by the context of the study. Given the lack of context in this study, our choices can be considered somewhat arbitrary.

Finally, it is worth noting that we are simulating perfect replications. Imperfect replications (i.e., those conducted in different laboratories, with different kinds of participants, under slightly different conditions) are useful for establishing generalizability. Here, we are purely interested in the ability of replications (in this case, artificial “perfect” replications) to control for multiplicity.

Results

Familywise Error Control

Tables 2 and 3 show that when studies are not replicated, the Bonferroni and Holm MCPs are, unsurprisingly, vastly preferable to no control. For example, Table 3 shows that in a non-replicated study with 100 participants and 21 comparisons ($\mu = 0,0,0,0,0,0,8$), the Bonferroni and Holm procedures keep the error rate below the nominal $\alpha = .05$. When no control is applied, the familywise error rate greatly exceeds $\alpha = .05$ (.374). In the more extreme situation where no true differences exist ($\mu = 0,0,0,0,0,0,0$), error rates are even worse when no control is used (.440).

When replications are conducted, error rates shift dramatically. Familywise error rates associated with the Bonferroni and Holm MCPs drop to .000, regardless of sample size and number of comparisons. Those associated with no control also drop noticeably. Tables 2 and 3 show that when a single replication is conducted, familywise error rates without multiplicity control are maintained at or below $\alpha = .05$. When two replications are conducted, empirical familywise error rates are maintained below .01.

Although we included results when both multiplicity control and replication are utilized, this paper specifically contrasts the use of multiplicity control in a single (non-replicated) study against a replicated study with no multiplicity control. Thus, the important contrast is between the Bonferroni and Holm results *without* replication, and that of the no control condition *with* replication. Tables 2 and 3 show that in this configuration, MCPs and no control with replication

both keep empirical familywise error rates at or below $\alpha = .05$. It is important to remind readers that replication was not designed, like MCPs, to maintain familywise error rates at specific levels. In fact, if the number of tests was very large it might take more than one or two replications to control the familywise error rate at α . The statistical properties of replication are nonetheless attractive—the probability of repeating an error over and over is very slim when the probability of an error in each instance (e.g., study) is small (i.e., α).

Power

A strategy that controls for familywise error by squandering statistical power has little utility. Therefore, we also compared the power provided by MCPs and replication. As expected, overall power (i.e., the probability of finding a significant effect in the original study and each replication) decreases as the number of replications increases, since the probability of finding a statistically significant effect across multiple replications is $(1 - \beta)^R$, where $1 - \beta$ is the power per replication, β is the Type II error rate, and R is the number of replications. Thus, with 2 replications and when a partial mean structure is used

($\mu = 0,0,0,8$ or $\mu = 0,0,0,0,0,8$), no control with one replication provided similar or higher per-pair power rates than multiplicity control without replication. Per-pair power rates for no control with two replications, and MCPs with no replications, are highly similar. When a complete non-null mean structure was used ($\mu = 0,8,16,24$ or $\mu = 0,8,16,24,32,40,48$) differences in per-pair power between replication and MCPs become slight (often inconsequential).

When all-pairs power was the outcome of interest, results were highly comparable except when sample sizes were large (e.g., $n = 100$) and the mean structure contained several true differences (e.g., $\mu = 0,8,16,24$ or $\mu = 0,8,16,24,32,40,48$). In these cases, the Holm procedure often demonstrated superior all-pairs power. For example, with population means of $0,8,16,24$ and $n = 100$, the Holm procedure had an all-pairs power rate of .450 whereas the all-pairs power

rate with no multiplicity control and one replication was .224 and with two replications was .104. The all-pairs power rate for the Bonferroni method was .089.

Meta-Analysis

As an extension of the previous results, we also explored the familywise Type I error and power rates when the results of the original study and the replication studies are combined into a single result. As expected, since no multiplicity control is imposed, familywise Type I error rates mirror what would be found in a single study with no multiplicity control. However, since meta-analysis combines the effects of multiple studies, the sample size, and hence the power, rises dramatically. Thus, both the per-pair and all-pairs power rates, especially with larger sample sizes, were much larger than any of the procedures where statistical significance is required in each of the studies conducted. Given recent support for the contention that Type I errors are theoretically implausible in most behavioral science research (e.g., Cribbie, 2017), focusing on power via meta-analytic solutions is very appealing.

Cohen's d

Effect sizes have become increasingly popular and are commonly used alongside traditional NHST. Unlike statistical significance, measures of effect size do not have statistical cut-offs. However, when there is little theoretical precedence for what constitutes a “meaningful” effect size, unofficial rules of thumb are often used to determine what constitutes a “minimal meaningful value”. One commonly used measure of effect size is Cohen's d , and $d = .3$ is often considered the cut-off for meaningful results.

Table 6 summarizes the ability of replication to prevent the effect size equivalent of a Type I error (obtaining a “meaningful” sample effect size when the population effect size is null) and Table 7 summarizes the ability of replication to prevent the effect size equivalent of a Type

II error (obtaining a sample effect size that is not meaningful when the population effect size is non-null). Recall that in our simulations we used $d = .3$ as the “minimal meaningful value”.

Table 6 shows that when sample sizes are sufficiently large ($n = 100$), replication effectively reduces the frequency of false conclusions about the meaningfulness of Cohen’s d . For example, when a partial-null mean structure was used (0,0,0,8 or 0,0,0,0,0,0,8), the probability that a Cohen’s d value was erroneously equal to or greater than $d = .3$ across all replications was less than 3% with a single replication, and approximately 0 with two.

Table 7 reports two different measures. The first is “Average Proportion of Correct” (APC) statements regarding the magnitude of Cohen’s d . This is the proportion of effect sizes in which Cohen’s d values were accurately at or above $d = .3$, averaged across all simulations. This shows how likely it is that truly meaningful effect sizes will persist across replications. The second is “Proportion of All Correct” (PAC) statements regarding the magnitude of Cohen’s d . This is the proportion of simulations in which *all* truly meaningful effect sizes were sufficiently large ($d \geq .3$) to be labelled as such. This shows how likely it is that every truly meaningful effect size in a study will persist across replications. These two measures are analogous to per-pair power and all-pairs power. Table 7 shows that APC and PAC “power rates” for Cohen’s d are highly comparable to our previously obtained NHST per-pair and all-pair power rates. Namely, rates decrease as the number of tests increase and the sample sizes decrease. This again highlights the importance of utilizing large sample sizes when possible.

Conclusion

Our simulation study yielded two important conclusions regarding the comparison of multiplicity control and replication on a statistical level. First, both the MCPs and replication maintained Type I error rates at acceptable levels. Second, replication and MCPs provide roughly equivalent power. We also extended the comparison by demonstrating that obvious

extensions of replication, such as focusing on effect sizes and meta-analyzing the results of replications, provide valuable research strategies. For example, meta-analysis, as expected, provides a huge advantage in power. Further, replication is a valuable strategy for minimizing the possibility that a researcher could incorrectly conclude that a meaningful effect size has been detected.

While there were some situations where replication performed better than multiplicity control and vice versa, the overall pattern suggested that MCPs and replication were very similar. Furthermore, our simulations show that when parameters most closely resembled those found in typical social science research studies (i.e., healthy sample size and a moderate number of comparisons where some but not all are truly different), replication provides satisfactory familywise error control and demonstrates equivalent or superior power. Thus, in most situations replication is either as good, or better, than multiplicity control.

While this paper directly compares replication and multiplicity control, it is not the case that researchers need to choose between the two. A decision regarding imposing multiplicity control is made independent of a decision regarding replication. However, combining the results of this study with the fact that replications are becoming commonplace in behavioural research provides researchers with valuable information on which to make a decision regarding the need for multiplicity control. Further, to reiterate early arguments, many of the assumptions underlying MCPs (e.g., null relationships are common), and subjective decisions required with MCPs (e.g., how to define an appropriate “family”), do not follow from common sense.

Replications are good scientific practice and should be always be conducted to ensure the generalizability and repeatability of one’s results, and to indirectly enable other beneficial research practices such as a comparison of effect sizes and a combining of effect sizes (i.e., meta-analysis). Without replicated studies, meta-analysis is simply not possible. The results of

this paper attempt to make the choice to conduct replications even more obvious by showing that replication, in addition to being theoretically superior to the strategy of MCPs, also provides a similar level of familywise error control. Accordingly, by adhering to best practices and replicating results, the use of MCPs becomes unnecessary. Further, our results highlight that conducting MCPs and replication together can severely limit the statistical power available to a researcher and is definitely not recommended.

In sum, the results of the present simulation study make the choice to conduct replications, and abandon the use of MCPs, even more obvious by demonstrating that, in addition to being theoretically superior, replication provides a similar level of familywise error control to MCPs. We hope this will encourage members of the social science community to take Wilkinson et al.'s shrewd advice to heart and "let replications promote reputations" (Wilkinson, 1990, p. 600).

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Appendix

Table 1

Simulation Mean Structure

	Familywise Error Control	Familywise Error Control/Power	Power
6 comparisons	$\mu = 0,0,0,0$ $ d_p = 0$	$\mu = 0,0,0,8$ $ d_p = 0$ or $.4^a$	$\mu = 0,8,16,24$ $ d_p = .4, .8, 1.2$
21 comparisons	$\mu = 0,0,0,0,0,0,0$ $ d_p = 0$	$\mu = 0,0,0,0,0,0,8$ $ d_p = 0$ or $.40$	$\mu = 0,8,16,24,32,40,48$ $ d_p = .4, .8, 1.2, 1.6, 2.0$ or 2.4

Note. d_p represents the population value for the pairwise Cohen's d ; when multiple d_p values are provided, e.g., for $\mu = 0,0,0,8$, $|d_p|$ can be 0 or .40, this implies that for some pairwise comparisons $|d_p| = 0$, e.g., for μ_1 vs μ_2 , $d_p = |(\mu_1 - \mu_2)/s_p| = |(0-0)/20| = 0$, and for other pairwise comparisons $|d_p| = .40$, e.g., for μ_1 vs μ_4 , $d_p = |(\mu_1 - \mu_4)/s_p| = |(0-8)/20| = |-.40| = .40$

Table 2

Familywise Error Rates for 4 groups ($T = 6$)

	n = 25		n = 100	
	$\mu = 0,0,0,0$	$\mu = 0,0,0,8$	$\mu = 0,0,0,0$	$\mu = 0,0,0,8$
No Replication				
Bonferroni	.041	.023	.038	.022
Holm	.041	.026	.038	.032
No Control	.202	.120	.205	.125
One Replication				
Bonferroni	.001	.000	.000	.000
Holm	.001	.000	.000	.000
No Control	.013	.007	.013	.007
Meta-analysis	.208	.120	.209	.121
Two Replications				
Bonferroni	.000	.000	.000	.000
Holm	.000	.000	.000	.000
No Control	.001	.000	.001	.000
Meta-analysis	.200	.117	.206	.122

Table 3

Familywise Error Rates for 7 groups (T = 21)

	n = 25		n = 100	
	$\mu = 0,0,0,0,0,0,0$	$\mu = 0,0,0,0,0,0,8$	$\mu = 0,0,0,0,0,0,0$	$\mu = 0,0,0,0,0,0,8$
No Replication				
Bonferroni	.039	.028	.043	.028
Holm	.039	.028	.043	.032
No Control	.442	.380	.440	.374
One Replication				
Bonferroni	.000	.000	.000	.000
Holm	.000	.000	.000	.000
No Control	.046	.036	.050	.031
Meta-analysis	.440	.363	.438	.362
Two Replications				
Bonferroni	.000	.000	.000	.000
Holm	.000	.000	.000	.000
No Control	.002	.002	.004	.002
Meta-analysis	.447	.371	.440	.364

Table 4

Average Per-Pair and All Pairs Power Rates for 4 Groups (6 Comparisons)

	n = 25				n = 100			
	$\mu = 0,0,0,8$		$\mu = 0,8,16,24$		$\mu = 0,0,0,8$		$\mu = 0,8,16,24$	
	PP	AP	PP	AP	PP	AP	PP	AP
No Replication								
Bonferroni	.103	.016	.380	.000	.566	.307	.782	.089
Holm	.110	.025	.419	.000	.596	.370	.885	.450
No Control	.288	.092	.567	.002	.802	.610	.903	.470
One Replication								
Bonferroni	.010	.000	.239	.000	.319	.096	.658	.009
Holm	.011	.001	.266	.000	.354	.139	.795	.203
No Control	.080	.008	.410	.000	.647	.374	.825	.224
Meta-analysis	.491	.241	.737	.042	.979	.948	.990	.941
Two Replications								
Bonferroni	.001	.000	.180	.000	.180	.032	.590	.001
Holm	.001	.000	.201	.000	.211	.053	.727	.090
No Control	.024	.000	.332	.000	.524	.233	.762	.104
Meta-analysis	.671	.433	.838	.215	.998	.995	.999	.996

Note. PP = Per Pair power rates, AP = All Pairs power rates

Table 5

Average Per-Pair and All Pairs Power Rates for 7 Groups (21 Comparisons)

	n = 25				n = 100			
	$\mu = 0,0,0,0,0,0,8$		$\mu = 0,8,16,24,32,40,48$		$\mu = 0,0,0,0,0,0,8$		$\mu = 0,8,16,24,32,40,48$	
	PP	AP	PP	AP	PP	AP	PP	AP
No Replication								
Bonferroni	.048	.001	.545	.000	.405	.083	.829	.000
Holm	.050	.002	.594	.000	.420	.100	.913	.150
No Control	.287	.039	.742	.000	.810	.487	.944	.202
One Replication								
Bonferroni	.002	.000	.450	.000	.160	.005	.758	.000
Holm	.002	.000	.494	.000	.172	.008	.851	.021
No Control	.080	.001	.642	.000	.645	.227	.898	.042
Meta-analysis	.501	.147	.907	.000	.977	.903	.994	.874
Two Replications								
Bonferroni	.000	.000	.407	.000	.064	.000	.729	.000
Holm	.000	.000	.448	.000	.071	.001	.809	.002
No Control	.022	.000	.592	.000	.515	.106	.862	.007
Meta-analysis	.682	.312	.890	.031	.998	.990	.999	.990

Note. PP = per pair power rates, AP = all pairs power rates

Table 6

Proportion of Incorrect Statements Regarding the Magnitude of Cohen's d for 4 and 7 Groups

	n = 25		n = 100	
	$\mu = 0,0,0,0$	$\mu = 0,0,0,8$	$\mu = 0,0,0,0$	$\mu = 0,0,0,8$
No Replication	.722	.543	.156	.090
One Replication	.372	.221	.007	.003
Two Replications	.132	.072	.000	.000
	$\mu = 0,0,0,0,0,0,0$	$\mu = 0,0,0,0,0,0,8$	$\mu = 0,0,0,0,0,0,0$	$\mu = 0,0,0,0,0,0,8$
No Replication	.938	.904	.347	.293
One Replication	.714	.612	.025	.016
Two Replications	.358	.275	.002	.000

Table 7

Average Proportion of Correct (APC) and Proportion of All Correct (PAC) Statements Regarding Magnitude of Cohen's d for 4 and 7 Groups

	n = 25				n = 100			
	$\mu = 0,0,0,8$		$\mu = 0,8,16,24$		$\mu = 0,0,0,8$		$\mu = 0,8,16,24$	
	APC	PAC	APC	PAC	APC	PAC	APC	PAC
No Replication	.651	.404	.810	.173	.758	.543	.880	.368
One Replication	.413	.155	.681	.032	.581	.301	.788	.137
Two Replications	.268	.063	.595	.005	.443	.168	.720	.051
	$\mu = 0,0,0,0,0,8$		$\mu = 0,8,16,24,32,40,48$		$\mu = 0,0,0,0,0,8$		$\mu = 0,8,16,24,32,40,48$	
	APC	PAC	APC	PAC	APC	PAC	APC	PAC
	APC	PAC	APC	PAC	APC	PAC	APC	PAC
No Replication	.647	.268	.890	.016	.765	.422	.931	.117
One Replication	.422	.070	.816	.000	.576	.170	.878	.016
Two Replications	.270	.015	.766	.000	.431	.065	.839	.002